Lecture 10

- Chain Rule (14.5)
- The Gradient Vector (14.6)

$f = f(x,y,z)$

$$\boxed{\Delta f \approx \underbrace{f_x}_{\frac{\partial f}{\partial x}(x,y,z)} \Delta x + f_y \Delta y + f_z \Delta z}$$

$x = x(t) \quad y = y(t) \quad z = z(t)$

$$\Delta f \approx f_x \underbrace{\frac{dx}{dt} \Delta t}_{\approx \Delta x} + f_y \frac{dy}{dt} \Delta t + f_z \frac{dz}{dt} \Delta t$$

Chain rule

$$\boxed{\frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}}$$

Example: $f = x^2 y + z$

$x = t \quad y = e^t \quad z = \sin t$

$$\frac{df}{dt} = 2xy \cdot 1 + x^2 \cdot e^t + 1 \cdot \cos(t)$$

$$= 2t e^t + t^2 e^t + \cos(t) //$$

Note: To know $\left.\frac{df}{dt}\right|_{t=1}$, need:

- f_x, f_y, f_z for $(x,y,z) = (1, e^1, \sin 1)$
- $\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}$ for $t = 1$.

Application: Product & quotient rules

$f = uv \qquad u = u(t), \quad v = v(t)$

$$\frac{d(uv)}{dt} = f_u \frac{du}{dt} + f_v \frac{dv}{dt}$$

$$= v \frac{du}{dt} + u \frac{dv}{dt}$$

$g = u/v \qquad u = u(t), \quad v = v(t)$

$$\frac{d(u/v)}{dt} = \frac{1}{v} \frac{du}{dt} + \left(\frac{-u}{v^2}\right) \frac{dv}{dt}$$

Chain rule w/ more variables

$f = f(x,y) \qquad x = x(u,v)$

$y = y(u,v)$

$$\frac{\partial f}{\partial u} = f_x \frac{\partial x}{\partial u} + f_y \frac{\partial y}{\partial u}$$

$$\frac{\partial f}{\partial v} = f_x \frac{\partial x}{\partial v} + f_y \frac{\partial y}{\partial v}$$

More work:

$$f_{rr} + \frac{1}{r} f_r + \frac{1}{r^2} f_{\theta\theta} = \overbrace{f_{xx} + f_{yy}}^{\text{Laplacian}}$$

<u>Example</u>: $x = r\cos\theta$

$y = r\sin\theta$

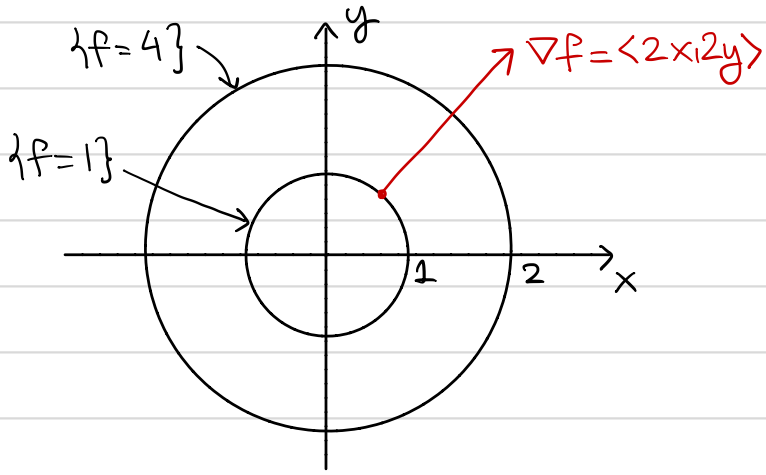
$$f_r = f_x \frac{\partial x}{\partial r} + f_y \frac{\partial y}{\partial r}$$

$$= f_x \cos\theta + f_y \sin\theta$$

$$f_\theta = f_x \frac{\partial x}{\partial \theta} + f_y \frac{\partial y}{\partial \theta}$$

$$= f_x(-r\sin\theta) + f_y\, r\cos\theta$$

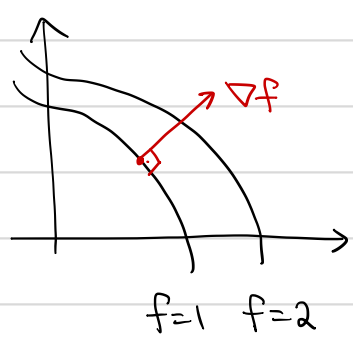
<u>Example</u>: $f(x,y) = x^2 + y^2$



<u>The gradient vector</u>

$$\boxed{\nabla f = \langle f_x, f_y, f_z \rangle}$$

<u>Theorem</u>: $\nabla f \perp$ level set $\{f = \text{constant}\}$



<u>Example</u>: $f = ax + by + cz$

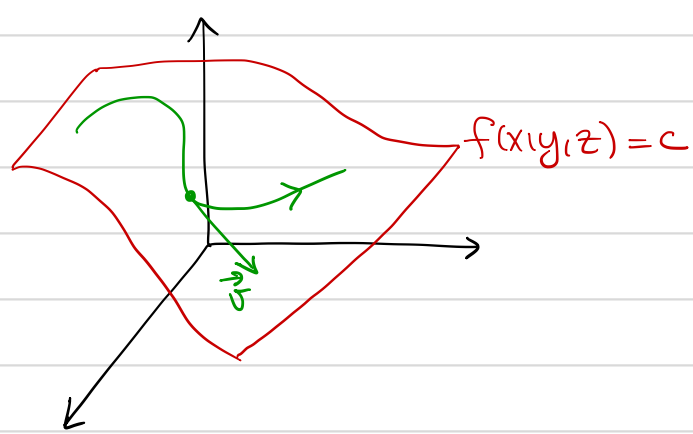
Level surface $\{\overbrace{ax+by+cz}^{\langle a,b,c\rangle\cdot\langle x,y,z\rangle} = d\}$ is plane with normal

$$\nabla f = \langle a, b, c \rangle.$$

$$\uparrow$$

$$\frac{\partial f}{\partial x}$$

<u>Proof</u>: Take a curve that stays at level $\{f = c\}$.



Curve: $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$

$\frac{d\vec{r}}{dt} = \vec{v}$ tangent to level surface

$$0 = \frac{df}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_z \frac{dz}{dt}$$

$$= \nabla f \cdot \vec{v}$$

Q.E.D.

<u>Application</u>: Find tangent plane of the surface

$$x^2+y^2-z^2=4$$

at $(2,1,1)$.

Level set $f=4$, where

$$f=x^2+y^2-z^2.$$

Gradient: normal

$$\nabla f = \langle 2x, 2y, -2z \rangle$$
$$= \langle 4, 2, -2 \rangle$$

Tangent plane: contains $(2,1,1)$

$$4x+2y-2z = 8$$